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RESEARCH ARTICLE

Are We Asking the Right Questions? Sex Specificity of Dream Characteristics in Probable Rapid Eye Movement Sleep Behavior Disorder

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Abstract: Background: Dream enactment behavior is a common feature of rapid eye movement (REM) sleep behavior disorder (RBD); however, there has been surprisingly little focus on sex differences in dream characterization in RBD. This is important to evaluate further, as it is possible that females and males have different dream characteristics in RBD, and screening questionnaires that focus on aggressive and injurious dream features (common to males) may be insensitive to detect RBD in female populations.
Objective: The aim was to determine if dream characteristics are sex specific in people with "probable" RBD (pRBD) and older adults from a large community cohort.
Methods: Adults aged ≥50 years were recruited from the Tasmanian community, Australia, to complete the REM Sleep Behavior Disorder Single-Question Screen (RBD1Q, to identify pRBD) and the Mannheim Dream Questionnaire (MADRE). Correlations between the MADRE and various outcomes were analyzed using multiple linear regression and χ^2 tests for binary response questions, with Bonferroni correction $P < 0.002$ denoting significance.
Results: One thousand eight hundred nine participants were recruited, and 7.1% (129) screened positive for pRBD. In the pRBD group, females reported more nightmares related to waking-life experiences ($P < 0.001$) than males. The pRBD group recalled dreams more often ($P < 0.001$) and experienced more lucid dreams ($P < 0.001$) and more nightmares during childhood ($P < 0.001$) than non-pRBD controls.
Conclusion: We found sex-specific differences in dream characteristics in pRBD and in pRBD versus controls. This is relevant to RBD screening questionnaires as questions on nightmares related to waking life may aid identification of females with pRBD.

Rapid eye movement (REM) sleep behavior disorder (RBD) is a parasomnia characterized by loss of normal atonia during REM sleep and dream enactment.[1] Isolated RBD (iRBD) is an early manifestation of α-synuclein-related neurodegenerative diseases (NDD), including Parkinson's disease (PD), dementia with Lewy bodies (DLB), and multiple system atrophy (MSA).[2] In contrast, "secondary" RBD develops in the presence of a neurological condition, including narcolepsy, autoimmune encephalitis, and

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brainstem lesions.[3–7] Identifying people with iRBD is important as approximately 80% will progress to an NDD within 14 years.[8]

Confirming the diagnosis of iRBD is challenging as it requires demonstration of REM sleep without atonia on a polysomnography (PSG), along with a clinical history of dream enactment behavior.[9] Screening questionnaires help identify "probable" RBD (pRBD), with specificity and sensitivity ranging between 36% and 98% and between 64% and 100%, respectively.[10,11] In 2021, a meta-analysis estimated the prevalence of PSG-confirmed iRBD to be about 0.68% in the general adult population, whereas pRBD was 5.65%.[12]

Several large-scale studies have found that males account for approximately 87% of RBD cases.[13,14] However, this striking sex specificity is not matched in clinically manifest NDDs, with a male-to-female ratio of 1.18 in PD[15] and almost 1:1 in DLB[16] and MSA.[17] Furthermore, several population-based studies have reported an equal sex split in PSG-confirmed RBD and iRBD.[18,19] Taken together, this suggests that females with iRBD are perhaps missed on traditional screening questionnaires. The reason for this could be that most screening questionnaires focus on aggressive dream enactment behaviors,[20,21] and there has been surprisingly little focus on sex differences in dream characterization. Therefore, it is unclear whether females have different dream characteristics in RBD.

In this study we investigated dream history in a large cohort (n = 1809) of males and females aged ≥50 years to determine if there were sex-specific differences in the dreams of those with pRBD compared to controls. To the best of our knowledge, this is the first study to focus on evaluating sex specificity of dream characteristics in pRBD. We hypothesized that dream characteristics would differ between males and females with pRBD, and between controls and people with pRBD. We addressed this through 2 aims:

i. To characterize dreaming in females and males with pRBD
ii. To characterize dreaming in people with pRBD compared to controls

Patients and Methods

Participants

Participants were recruited from the general population of Tasmania, an island state in Australia. Eligibility criteria were (1) being a resident of Tasmania, (2) being aged ≥50 years, and (3) being a participant in the Island Study Linking Aging and Neurodegenerative Disease (ISLAND) Project. The ISLAND Project is a public health initiative launched in June 2019 at the University of Tasmania, aiming to decrease dementia risk in Tasmanians (a previously published protocol).[22] Methods of recruitment for the present study included community poster/leaflet advertisements placed in hospitals, supermarkets, and community groups, and email invitations to participants who were already enrolled in the ISLAND Sleep Study project,[23] a substudy of the ISLAND Project.[22] Participants were required to enroll in the ISLAND Project if recruited from the community to allow for data collaboration between the two research projects.

The ISLAND Sleep Study was approved by the University of Tasmania Human Research Ethics Committee (REF26435), and this project was carried out according to the National Statement on Ethical Conduct in Human Research (2022). Participants provided consent online.[22,23]

Data Collection

Participants completed a self-paced battery of online health and sleep questionnaires between June and August of 2022, presented in a fixed order, and they could choose which questionnaires they completed (see Table S1). To identify pRBD we used the REM Sleep Behavior Disorder Single-Question Screen (RBD1Q) that simply asks "Have you ever been told, or suspected yourself, that you seem to 'act out your dreams' while asleep (for example, punching, flailing your arms in the air, making running movements, etc.)?"[24] It has a sensitivity/specificity of 80%/75.3% for video polysomnography (vPSG)–confirmed RBD in older adult populations,[10] making it one of the most reliable screening measures for RBD.[11] Participants screening positive were defined as pRBD and those who screened negative as controls.

The self-report Mannheim Dream Questionnaire (MADRE)[25] was used to investigate dream characteristics; it collects information on dream recall, intensity, and emotional tone, as well as nightmares, lucid dreaming, attitude toward dreams, telling dreams, recording dreams, dreams affecting daytime mood, creative dreams, problem-solving dreams, and déjà vu experiences. Questions include "How intense are your dreams emotionally?," "What is the emotional tone of your dreams on average?," "How often have you experienced nightmares recently (in the past several months)?," and "How often do you experience so-called lucid dreams (when you know that you are dreaming while you are asleep)?". Dream recall questions were presented on a 7-point Likert scale (0 = never, 1 = less than once a month, 2 = about once a month, 3 = about 2–3 times a month, 4 = about once a week, 5 = several times a week, 6 = almost every morning). Emotional dream intensity is presented on a 5-point scale (0 = not at all intense, 1 = not that intense, 2 = somewhat intense, 3 = quite intense, 4 = very intense), as is emotional tone (−2 = very negative, − 1 = somewhat negative, 0 = neutral, +1 = somewhat positive, +2 = very positive). Question 9 "Please list the topics of your childhood nightmares" was not analyzed for this paper due to the qualitative nature of the question. All other question responses are scored on a 7- to 8-point Likert scale or a binary yes/no option, specific to each question.[25]

To provide more details on this cohort, participants also completed the Baylor Functional Assessment Scale (BFAS),[26] which comprises 12 yes/no questions about movement changes validated to differentiate PD from controls.[26] The Composite Autonomic Symptom Score-31 (COMPASS-31) [27], which has good correlation with objective autonomic test results in people with PD,[28] was also administered.

In addition, cognitive function data were extracted from the ISLAND annual assessments (completed online in October 2021) in a subsample of participants (n = 1292) who completed the paired associates learning (PAL, measure of episodic memory) and spatial working memory (SWM, measure of executive function) tasks of the Cambridge Neuropsychological Test Automated Battery[29], previously validated to detect cognitive changes in α-synucleinopathies.[29–31]

Olfactory dysfunction was also evaluated using University of Pennsylvania Smell Identification Test (UPSIT) kits in a subsample of participants (n = 228) from the pRBD and healthy control groups. Kits were posted out and returned via prepaid envelopes between February and December 2023. The UPSIT is a 40-item self-administered "scratch and sniff" test that requires participants to select the correct odor from 4 options. It is well validated for research, and age- and sex-specific normative data have been published.[32]

Additional data were extracted from the ISLAND Project, including age (years), sex (male/female), marital status (single, married, de facto, separated or divorced, widowed), level of education (primary school, high school, certificate or apprenticeship, diploma/associates degree, bachelor's degree, higher university degree, other), employed (yes/no), retired (yes/no), remoteness area (inner regional Australia, outer regional Australia, remote Australia, very remote Australia), and the Hospital Anxiety and Depression Scale (HADS)[33] (normal, borderline abnormal, abnormal; see Appendix 1).

Data Analysis

Numerical data were expressed as mean and standard deviation (SD), and categorical variables as number and percentage. Correlations between the MADRE questions and various outcomes, including confounders, were analyzed using multiple linear regression with post hoc corrections to control for type 1 error rate, as well as χ^2 tests for binary response questions. Differences in sexes in the pRBD groups were determined by estimating the interaction between sex and pRBD, adjusted for covariates. Confounders of obstructive sleep apnea (OSA), anxiety, and depression were used in the linear regression model. OSA was measured using the snoring, tiredness, observed apnea, high blood pressure, body mass index, age, neck circumference, and male gender (STOP-Bang) questionnaire (see Appendix 2).[34] This was chosen as a confounder as descriptive statistics revealed that those with pRBD had a higher risk for OSA (see Table 2), and OSA is known to mimic RBD symptoms.[35] Anxiety and depression scores on the HADS were higher in the pRBD group and thus may confound results of emotional-based questions. Regular antidepressant usage was higher in the pRBD group, but it was not used as a confounder due to the potential bidirectional relationship between antidepressant use and iRBD.[36] Based on the Bonferroni correction (for 25 individual MADRE questions), a P-value <0.002 denoted statistical significance. All statistical analyses were performed using R-studio.[37]

Results

Description of pRBD and Control Cohorts

Over 4000 ISLAND Project participants, as well as additional community member recruitment through advertisements, were invited to take part in the Tasmanian ISLAND Sleep Study. One thousand eight hundred nine completed the RBD1Q and MADRE, with 129 (7.1%) screening positive on the RBD1Q (mean [SD] age: 62 [7.7] years, 58% female) and 1673 screening negative (mean [SD] age: 64 [7.6] years, 79% female). Demographic data stratified by pRBD status are presented in Table 1.

COMPASS scores were similar between groups, whereas BFAS scores were found to be higher in females in the pRBD group. Cognitive scores showed slightly less impairment on both SWM and PAL tasks in the pRBD groups compared to controls, and olfactory scores were also found to be lower in the female pRBD group compared to the female control group (see Table 2). Results of MADRE questions within and between groups are discussed later and presented in Figure 1.

Dream Characteristics between Males and Females in the pRBD Group

In the pRBD group, females experienced more nightmares related to experiences in waking life compared to males (35% vs. 14%, $\chi^2 = 19.39$; $P < 0.001$). They also reported that dreams affected their mood more during the day (mean [SD] = 2.39 [2.25] vs. 0.944 [1.57], $F = 14.31$; $P < 0.001$); however, this was also a significant difference between pRBD and controls (1.78 [2.11] vs. 0.936 [1.43], $F = 41.75$; $P < 0.001$). No significant differences were found between males and females with pRBD for any other MADRE questions (see Table S2).

Dream Characteristics between Control and pRBD Groups

Dream recall was higher in those with pRBD compared to controls (3.78 [1.72] vs. 2.85 [1.70], $F = 36.6528$; $P < 0.001$). Nightmares were a key feature of pRBD: those with pRBD experienced more recent (2.95 [1.87] vs. 1.65 [1.67], $F = 76.51$; $P < 0.001$) and distressing (1.42 [1.10] vs. 0.969 [0.953], $F = 22.72$; $P < 0.001$) nightmares than controls and reported that they experienced more nightmares during childhood (3.34 [2.27] vs. 2.39 [2.14], $F = 23.60$; $P < 0.001$). Additionally, compared to controls, participants with pRBD reported more emotionally intense dreams (1.84 [1.04] vs. 1.21 [0.973], $F = 52.40$; $P < 0.001$), experienced more lucid dreams (2.51 [2.18] vs. 1.69 [1.96], $F = 20.41$; $P < 0.001$) and déjà vu (2.08 [1.68] vs. 1.46 [1.39], $F = 23.43$; $P < 0.001$), attributed more meaning to their dreams (1.60 [1.16] vs. 1.28 [1.08], $F = 11.63$; $P < 0.001$), wanted to know more about dreams (2.02 [1.37] vs. 1.60 [1.31],

TABLE 1 Demographic variables stratified by RBD1Q response and sex

	Control (N = 1673; 93%)		pRBD (N = 129; 7%)	
	Female (N = 1323; 79%)	Male (N = 350; 21%)	Female (N = 75; 58%)	Male (N = 54; 41%)
Age, N (%)				
Mean (SD)	63.5 (7.33)	66.9 (7.97)	59.9 (6.31)	64.6 (8.61)
Median [min, max]	63.0 [50.0, 86.0]	67.5 [50.0, 91.0]	59.0 [50.0, 80.0]	64.0 [50.0, 88.0]
Marital status, N (%)				
Married	760 (57.4%)	242 (69.1%)	48 (64.0%)	44 (81.5%)
Separated or divorced	193 (14.6%)	34 (9.7%)	11 (14.7%)	2 (3.7%)
De facto	127 (9.6%)	34 (9.7%)	7 (9.3%)	5 (9.3%)
Single	107 (8.1%)	21 (6.0%)	4 (5.3%)	1 (1.9%)
Widowed	107 (8.1%)	12 (3.4%)	4 (5.3%)	2 (3.7%)
Other	9 (0.7%)	0 (0%)	1 (1.3%)	0 (0%)
Prefer not to say	4 (0.3%)	0 (0%)	0 (0%)	0 (0%)
Missing	16 (1.2%)	7 (2.0%)	0 (0%)	0 (0%)
Highest education level, N (%)				
Higher university degree (honors, graduate diploma, master's, or PhD)	438 (33.1%)	121 (34.6%)	28 (37.3%)	27 (50.0%)
Bachelor's degree	299 (22.6%)	71 (20.3%)	12 (16.0%)	11 (20.4%)
Diploma/associate degree	248 (18.7%)	60 (17.1%)	11 (14.7%)	8 (14.8%)
Certificate or apprenticeship (including Cert 2, 3, or 4)	115 (8.7%)	44 (12.6%)	12 (16.0%)	4 (7.4%)
High school	163 (12.3%)	43 (12.3%)	9 (12.0%)	4 (7.4%)
Other	47 (3.6%)	7 (2.0%)	3 (4.0%)	0 (0%)
Missing	13 (1.0%)	4 (1.1%)	0 (0%)	0 (0%)
Employed, N (%)				
No	769 (58.1%)	237 (67.7%)	37 (49.3%)	36 (66.7%)
Yes	546 (41.3%)	109 (31.1%)	37 (49.3%)	18 (33.3%)
Missing	8 (0.6%)	4 (1.1%)	1 (1.3%)	0 (0%)
Retired, N (%)				
No	109 (8.2%)	22 (6.3%)	5 (6.7%)	6 (11.1%)
Yes	676 (51.1%)	217 (62.0%)	33 (44.0%)	31 (57.4%)
Missing	538 (40.7%)	111 (31.7%)	37 (49.3%)	17 (31.5%)
Remoteness area[a], N (%)				
Inner regional Australia	972 (73.5%)	259 (74.0%)	56 (74.7%)	45 (83.3%)
Outer regional Australia	334 (25.2%)	87 (24.9%)	18 (24.0%)	9 (16.7%)
Remote Australia	3 (0.2%)	1 (0.3%)	0 (0%)	0 (0%)
Very remote Australia	3 (0.2%)	1 (0.3%)	0 (0%)	0 (0%)
Missing	11 (0.8%)	2 (0.6%)	1 (1.3%)	0 (0%)

[a]Relative geographic remoteness in Australia is measured by calculating road distance from various populated locations.[66]

Abbreviations: RBD1Q, Rapid Eye Movement Sleep Behavior Disorder Single-Question Screen; pRBD, probable rapid eye movement sleep behavior disorder; SD, standard deviation; min, minimum; max, maximum.

TABLE 2 Clinical features stratified by RBD1Q response and sex

	Control (N = 1673; 93%)		pRBD (N = 129; 7%)	
	Female (N = 1323; 79%)	Male (N = 350; 21%)	Female (N = 75; 58%)	Male (N = 54; 41%)
Risk of OSA, N (%)				
High risk of OSA	2 (0.2%)	9 (2.6%)	0 (0%)	3 (5.6%)
Intermediate risk of OSA	112 (8.5%)	192 (54.9%)	9 (12.0%)	31 (57.4%)
Low risk of OSA	1209 (91.4%)	149 (42.6%)	66 (88.0%)	20 (37.0%)
Anxiety category, N (%)				
Abnormal	73 (5.5%)	11 (3.1%)	12 (16.0%)	4 (7.4%)
Borderline abnormal	124 (9.4%)	20 (5.7%)	12 (16.0%)	7 (13.0%)
Normal	1126 (85.1%)	319 (91.1%)	51 (68.0%)	43 (79.6%)
Depression category, N (%)				
Abnormal	18 (1.4%)	3 (0.9%)	7 (9.3%)	2 (3.7%)
Borderline abnormal	52 (3.9%)	10 (2.9%)	7 (9.3%)	2 (3.7%)
Normal	1253 (94.7%)	337 (96.3%)	61 (81.3%)	50 (92.6%)
Regular antidepressant use, N (%)				
No	1132 (85.6%)	316 (90.3%)	52 (69.3%)	42 (77.8%)
Yes	189 (14.3%)	34 (9.7%)	22 (29.3%)	12 (22.2%)
Missing	2 (0.2%)	0 (0%)	1 (1.3%)	0 (0%)
Olfactory score				
Mean (SD)	31.0 (3.88)	27.7 (4.85)	29.3 (5.27)	28.5 (5.46)
Median [min, max]	31.0 [13.0, 37.0]	29.0 [17.0, 36.0]	30.0 [12.0, 36.0]	30.0 [11.0, 37.0]
Missing	1215 (91.8%)	309 (88.3%)	27 (36.0%)	28 (51.9%)
COMPASS-31 total (Autonomic Symptom Score)				
Mean (SD)	14.7 (10.2)	14.5 (10.1)	14.6 (9.90)	13.5 (8.24)
Median [min, max]	13.4 [0, 56.5]	13.3 [0, 50.2]	13.6 [0.666, 40.8]	14.1 [0.666, 30.2]
Missing	135 (10.2%)	27 (7.7%)	9 (12.0%)	6 (11.1%)
BFAS-adapted total (parkinsonism symptom score)				
Mean (SD)	1.69 (1.73)	1.97 (1.79)	2.09 (2.05)	1.94 (2.17)
Median [min, max]	1.00 [0, 11.0]	2.00 [0, 11.0]	2.00 [0, 9.00]	1.00 [0, 11.0]
CANTAB SWM (between errors 6-pattern)				
Mean (SD)	3.48 (3.35)	3.15 (3.42)	2.81 (2.87)	2.72 (3.33)
Median [min, max]	3.00 [0, 14.0]	2.00 [0, 12.0]	2.00 [0, 9.00]	0 [0, 10.0]
Missing	369 (27.9%)	97 (27.7%)	32 (42.7%)	15 (27.8%)
CANTAB SWM (total errors 6-pattern)				
Mean (SD)	3.58 (3.44)	3.24 (3.50)	2.95 (3.08)	2.74 (3.35)
Median [min, max]	3.00 [0, 17.0]	2.00 [0, 13.0]	2.00 [0, 10.0]	0 [0, 10.0]
Missing	369 (27.9%)	97 (27.7%)	32 (42.7%)	15 (27.8%)

(Continues)

TABLE 2 *Continued*

	Control (N = 1673; 93%)		pRBD (N = 129; 7%)	
	Female (N = 1323; 79%)	Male (N = 350; 21%)	Female (N = 75; 58%)	Male (N = 54; 41%)
CANTAB PAL (total errors 6-pattern adjusted)				
Mean (SD)	2.28 (1.10)	2.41 (1.16)	2.21 (1.06)	2.38 (1.11)
Median [min, max]	2.00 [0, 4.00]	2.00 [0, 4.00]	2.00 [0, 4.00]	2.00 [1.00, 4.00]
Missing	365 (27.6%)	98 (28.0%)	32 (42.7%)	15 (27.8%)
CANTAB PAL (total errors 6-pattern)				
Mean (SD)	3.78 (4.07)	4.48 (4.53)	3.56 (3.74)	3.92 (4.47)
Median [min, max]	3.00 [0, 22.0]	3.00 [0, 21.0]	2.00 [0, 14.0]	2.00 [0, 18.0]
Missing	365 (27.6%)	98 (28.0%)	32 (42.7%)	15 (27.8%)

Abbreviations: RBD1Q, Rapid Eye Movement Sleep Behavior Disorder Single-Question Screen; pRBD, probable rapid eye movement sleep behavior disorder; OSA, obstructive sleep apnea; SD, standard deviation; min, minimum; max, maximum; COMPASS-31, Composite Autonomic Symptom Score-31; BFAS, Baylor Functional Assessment Scale; CANTAB, Cambridge Neuropsychological Test Automated Battery; SWM, spatial working memory; PAL, paired associates learning.

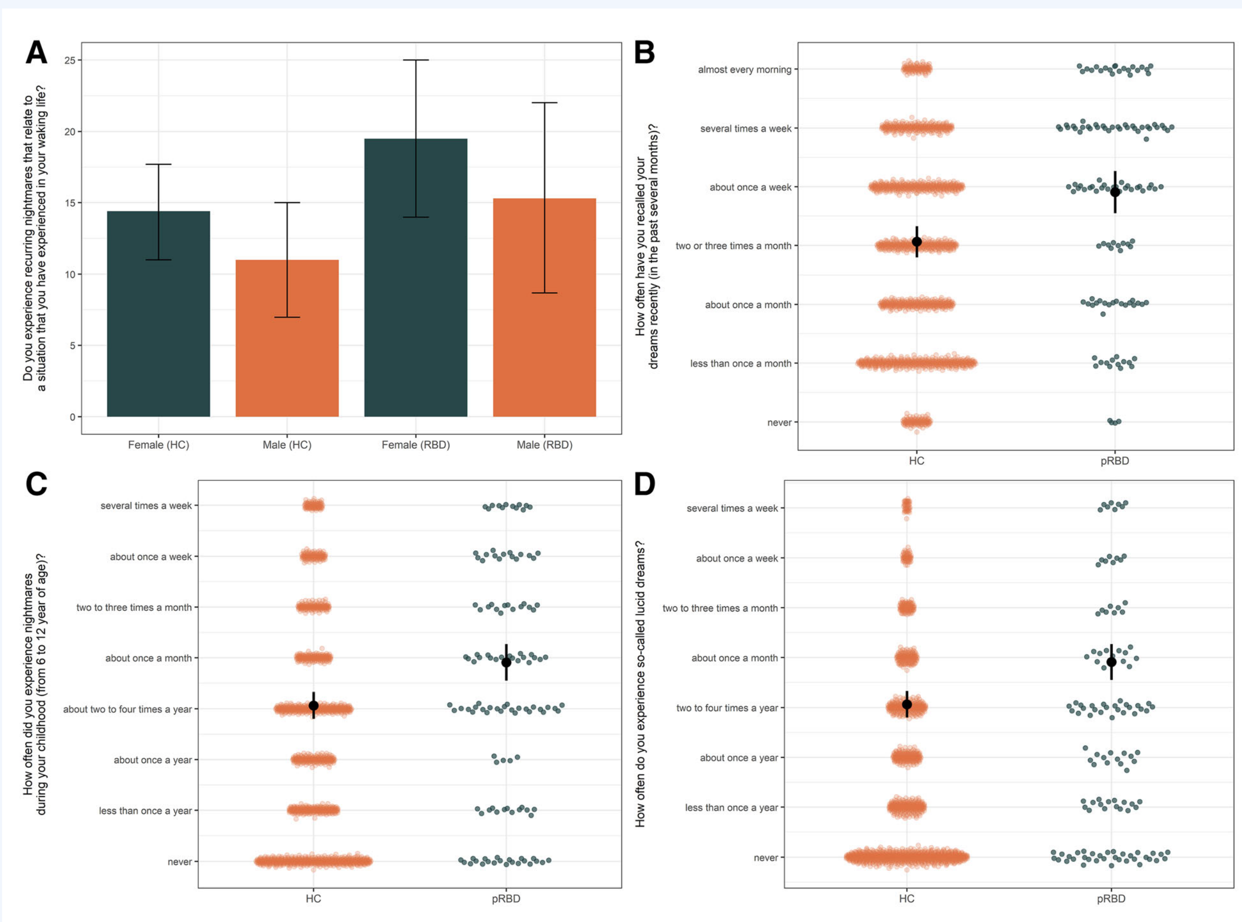

FIG. 1. Key findings within and between groups. Females with pRBD (probable rapid eye movement sleep behavior disorder) reported more nightmares related to (**A**) waking life compared to pRBD males. The pRBD group (**B**) recalled their dreams more often, (**C**) experienced more nightmares in childhood, and (**D**) reported more episodes of lucid dreaming than controls. Colored points, question responses; black points, means; black whiskers, 95% confidence intervals.

$F = 12.72$; $P < 0.001$), told their dreams to others more often (3.10 [2.13] vs. 2.26 [1.75], $F = 27.22$; $P < 0.001$), and were given more creative ideas by their dreams (1.36 [1.72] vs. 0.877 [1.43], $F = 13.21$; $P < 0.001$).

No significant difference in recurrent nightmares, life enrichment from dreams, frequency of dream recording, or problem-solving from dreams was found between the pRBD and controls groups. These features also did not differ between males and females (see Table S2).

Discussion

This is the first study to investigate sex differences in dreaming in pRBD and one of few studies to analyze dream characteristics in a large cohort of healthy community-based older adults. We found evidence to support both our hypotheses: dream characteristics differed between males and females with pRBD, and between controls and people with pRBD. Females with pRBD reported a higher frequency of nightmares related to experiences in waking life compared to males with pRBD, and those with pRBD recalled their dreams more often, experienced more nightmares in childhood and more lucid dreaming, told their dreams more often to others, and were given more creative ideas from their dreams, than controls.

Current RBD screening questionnaires ask about dream enactment,[10] but most tend to focus on vivid dreams with aggressive, defensive, or action-packed content,[38,39] rather than dream recall or nightmares. This study, and the strong male predominance in iRBD, raises questions on whether these types of "aggressive, defensive, or action-packed" dreams are a central feature of *male* RBD dreams rather than RBD in general.[40] This is important to clarify as current screening methods tend to overlook certain characteristics of dreaming which may be more sensitive at detecting RBD in females.

Dreaming in Females versus Males with pRBD

Our finding that females with pRBD report more nightmares related to experiences in waking life is supported by previous evidence showing that females in general tend to report more health- and death-related nightmares, including death/injury of others, as well as more nightmares of sexual aggression.[41] Females tend to engage in these themes, which are particular to social relationships, more often and more intensely, than males do.[42] Waking-life stressors are known to trigger the occurrence of nightmares,[43] and post-traumatic stress disorder has been associated with the development of PD,[44] so it may be that females with RBD are exposed to greater life stress and thus experience increased nightmares related to waking experiences in prodromal stages of NDD. However, further research is needed to determine whether stress and trauma are risk factors for iRBD. Nevertheless, these findings indicate that nightmare content should be further explored in RBD screening questionnaires as it was a prominent feature of female pRBD dreams, outside of the usual topics of action and aggression, and this novel finding has not been considered as a key symptom of RBD before.

Our results of frequent dreams and distressing nightmares in pRBD accord with previous literature that has found that vivid and aggressive dreams are a prominent feature of RBD.[20,21] However, most studies included samples with significantly greater numbers of males than females.[45] A recent narrative review investigated dream activity in PSG-confirmed iRBD and RBD, and found that most participants reported recurrent unpleasant dreams, often with aggressive and violent content.[45] Analysis of the male-to-female ratio in 26 studies included in this review shows that only 2 had a sample of RBD participants with an even number of males and females, both with very small sample sizes (n < 6). Notably, 92% of studies included a significantly greater number of males in their samples, indicating that dream characteristics in females with RBD have been underrepresented in the scientific literature.[45]

Dreaming in People with pRBD versus Controls

The pRBD group reported a higher incidence of dream recall than controls and experienced more nightmares in childhood and more lucid dreams. These findings align with previous research; for example, one study including 162 participants found that dream recall frequency was significantly higher in those with RBD than controls (77% vs. 35%),[46] and another study involving 68 RBD and 44 control participants also found RBD participants remembered more dreams and nightmares compared to controls.[47]

This heightened dream recall in pRBD could potentially be attributed to the neurobiological alterations associated with RBD,[48] which might influence the transition between sleep stages and dream memory consolidation. Several theories of dream recall in sleep disorders have been posited, including the "arousal–retrieval" model of dream recall, the "continuity hypothesis of dreaming," and the body–mind interaction during REM sleep.[49] The arousal–retrieval model describes 2 necessary steps for the retrieval of dreams: first, cortical arousal to transfer information from short-term storage to long-term memory, and for this to happen, a period of wakefulness needs to occur, as it is proposed that this storage transfer cannot occur during sleep.[49] In the case of RBD, it may be that the frequent nighttime arousals associated with RBD episodes increase the transfer of dream content to long-term memory, which could explain the increase in their dream recall compared to controls who experience less awakenings overnight. Similarly, Fasiello and colleagues suggested that "recall-bias syndrome" may explain the tendency for people with RBD to recall dreams more frequently due to their dream-related awakenings.[50] It is also possible that the greater attribution of meaning to dreams leads to more frequent dream recall in pRBD than in controls, as their heightened awareness of dream experiences primes them for greater recall. The telling of dreams could also be a form of emotional processing as dreams often serve as a platform for emotional expression,[51] and individuals with pRBD may find solace in exploring their dreams. It should be noted, however, that those in the pRBD group did report greater regular use of antidepressant medication than controls; this is relevant as antidepressant usage has previously been associated with more intense dreams and nightmares.[52] Importantly though, our current understanding of the antidepressant effect on REM sleep includes their potential to unmask iRBD.[53] But although there may be a bidirectional relationship between antidepressant use and PSG-confirmed iRBD, it is unclear from the present data whether regular antidepressant use could have influenced the observed sex differences found in dreaming in this study.

A novel finding was that lucid dreaming is experienced more by those with pRBD; a phenomenon whereby one is aware that he or she is dreaming while asleep and may be able to consciously alter the content of the dream.[54] It occurs most commonly in late-night REM sleep and is associated with increased physiological activation during REM sleep: specifically, elevated levels of eye muscle activity (or REMs) have been shown to occur during lucid dreaming episodes in REM sleep.[55] Lucid dreaming occurs in 77% of people with narcolepsy (a sleep disorder characterized by excessive daytime sleepiness and rapid entry into REM sleep),[56] and 60% of people with narcolepsy also have RBD.[5,56,57] It may be that people with narcolepsy and RBD share similar neurobiological changes such as dysfunctions in the hypocretin/dopaminergic systems,[57] as both these systems are known to be involved in the pathology of narcolepsy and PD.[58,59] Furthermore, lucid dreaming should be explored as a possible therapeutic tool for those with RBD, as Holzinger and colleagues have found lucid dreaming to be an effective intervention for those experiencing nightmares and post-traumatic stress disorder alongside anxiety and depression.[60]

Intriguingly, and for the first time, we found that participants with pRBD reported more nightmares in childhood than controls. Research shows that childhood trauma and adversity are associated with nightmares in adulthood[61] and that there is an association between distressing dreams in childhood and the development of cognitive impairment or PD in later life.[62] It is unclear if this is related to stress–response mechanisms, raising the risk of neurodegenerative disorders,[63] and further research in this field is required. The finding should, however, be interpreted cautiously. Given that the sample includes probable (not confirmed) iRBD, and potentially includes non-REM parasomnia cases, this retrospective association should be appropriately tempered.

Strengths, Limitations, and Future Directions

The key strengths of this study are the large cohort of older adults with a broad range of ages from 50 to 90 years and the sizable pRBD sample (n = 129) of whom 58% are female, which is a noteworthy and underrepresented feature in RBD research. A wealth of additional data and variables to fully understand potential confounding effects in this cohort is robust and enabled the adjustments for significant factors of OSA, as well as anxiety and depressive symptoms.

Several limitations are acknowledged; the original sample of the ISLAND Sleep Study was female predominant and highly educated, and prone to self-selection "healthy bias." This benefited the near even number of males and females of our pRBD sample but also means that the cohort is not representative of the general population. Tasmania itself is not representative of the broader cultural diversity in Australia either, with the majority of its population White and of northern European ancestries. Furthermore, the term "sex" has been used to categorize participants, but it is possible that the reported sex of male/female is not concordant with participant's biological sex and instead has been answered for their gender identity. OSA risk was also found to be higher in the pRBD group, and as such this was accounted for in the analyses. Conversely, cognitive testing revealed slightly lower scores in the pRBD group, which may have an influence on dream experiences and dream recall,[64] and further work is underway to better characterize and confirm iRBD within this cohort.

In addition, the reliance on self-report measures introduces recall bias and did not adjust for the time of day that questions were completed. Also, the use of questionnaires may not be suitable for older people when subtle cognitive decline may be emerging, particularly for capturing potential mediators such as OSA.[65] Furthermore, we did not measure lifetime diagnoses of anxiety, major depression, post-traumatic stress disorder, or insomnia, which all have the potential to impact results.

We acknowledge that reliance on a screening questionnaire for the pRBD group is a limitation to this study, with the possibility of some participants screening false positive on the RBD1Q, due to other sleep disorders such as OSA or non-REM parasomnias. The absence of bed partner confirmation of dream enactment behavior also introduces potential bias, such that those without a bed partner, and especially those who do not frequently recall their dreams, may be more likely to screen negative for RBD. This study is ongoing, and home-based vPSGs are currently underway to confirm the presence of RBD.

Conclusion

This study has explored the intricate relationship between sex differences in dream characteristics and pRBD. Key findings are that dream recall, childhood nightmares, and lucid dreaming are significant features of pRBD. For females with pRBD, nightmares relating to waking events are also a prominent characteristic. Our research suggests that it may be time to rethink how we screen for RBD, with a focus on the female experience. It points toward the potential utility of introducing questions on dream recall and dream content, particularly in reference to nightmares related to waking life; this may aid identification of females with RBD in the community, but further research in vPSG-confirmed iRBD groups is needed to strengthen this recommendation. This study also raises the question of whether females with iRBD present for medical investigation less often than males, as their dreams tend to be less aggressive and injurious, which may contribute to a male predominance in iRBD diagnosis. As such, those screening for iRBD need to consider sex-specific dream differences—not only for the management of iRBD, but also in preparation for clinical trials investigating neuroprotective medications in these high-risk patients.

Author Roles

(1) Research project: A. Conception, B. Organization, C. Execution; (2) Statistical analysis: A. Design, B. Execution, C. Review and critique; (3) Manuscript preparation: A. Writing of the first draft, B. Review and critique.

S.B.: 1A, 1B, 1C, 2A, 2B, 2C, 3A, 3B
A.J.N.: 1A, 2C, 3B
L.P.-C.: 2C, 3B
A.E.K.: 1A, 2C, 3B
S.L.N.: 1B, 2C, 3B
X.W.: 2B, 3B
A.B.: 2A, 2B, 2C, 3B
J.C.V.: 1A, 3B
J.A.: 1A, 1B, 2C, 3A, 3B

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Disclosures

Ethical Compliance Statement: The University of Tasmania Human Research Ethics Committee reviewed and approved this study. All participants in this study gave their written informed consent via an online consent form prior to the inclusion of their data in the study. All identifying information was removed prior to analysis. We confirm that we have read the journal's position on issues involved in ethical publication and affirm that this work is consistent with those guidelines.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. ■

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Appendix 1

Scoring criteria for Hospital Anxiety and Depression Scale (HADS)

Anxiety:

- Normal: scores below the cutoff point
- Borderline abnormal: scores falling within the borderline range
- Abnormal: scores above the cutoff point
- Cutoff scores:
- Normal: HADS-A score <8
- Borderline abnormal: 8 ≤ HADS-A score ≤ 10
- Abnormal: HADS-A score > 10

Depression:

- Normal: scores below the cutoff point
- Borderline abnormal: scores falling within the borderline range
- Abnormal: scores above the cutoff point
- Cutoff scores:
- Normal: HADS-D score <8
- Borderline abnormal: 8 ≤ HADS-D score ≤ 10
- Abnormal: HADS-D score > 10

Appendix 2

Scoring criteria for STOP-Bang

Low risk of OSA: yes to 0–2 questions.

Intermediate risk of OSA: yes to 3–4 questions.

High risk of OSA: yes to 5–8 questions or yes to 2 or more of 4 STOP questions + male gender or yes to 2 or more of 4 STOP questions + BMI > 35 kg/m^2 or yes to 2 or more of 4 STOP questions + neck circumference (17″/43 cm in males, 16″/41 cm in females).

Supporting Information

Supporting information may be found in the online version of this article.

Table S1. Full details of all online health and sleep questionnaires completed by the ISLAND (Island Study Linking Aging and Neurodegenerative Disease) Sleep Study cohort.

Table S2. Statistical correlation results between MADRE (Mannheim Dream Questionnaire) questions and all groups. This details the significance of each question within the pRBD (probable rapid eye movement sleep behavior disorder) group, between pRBD and controls, and also within the control group.